13. Dashboard Design and Interactions

Purpose

Provide an overview of the supplied EV dataset and allow users to compare recorded attributes without implying that record counts equal market shares.

Recommended layout

1. **Header:** project title, file coverage, and source caveat.
2. **KPI row:** records, manufacturers, categories, regions, mean/median price, median range.
3. **Composition row:** category counts and region counts.
4. **Comparison row:** median price and median range by category.
5. **Technical row:** battery capacity vs range; price vs battery capacity.
6. **Distribution row:** efficiency, sales, ratings, or charging distributions.
7. **Footer:** definitions, source/provenance caveat, last profile date.

Filters

- Model year
- Market region
- Manufacturing country
- Vehicle category
- Battery type
- Drivetrain
- Recommended use case

Only add filters that work consistently across the selected worksheets. Provide a clear reset instruction and show active filter context.

Interactions

- Selecting a category may highlight related points in comparison charts.
- Tooltips should show category, metric, units, and count.
- Avoid excessive cross-filtering that makes the dashboard hard to interpret.
- Consider excluding missing values only for the metric that requires them.

Validation

Compare KPI values with the baseline values in the KPI document. Check that filtering changes both the number and metric consistently. Confirm axis titles, units, and color legends are readable in the published embed.